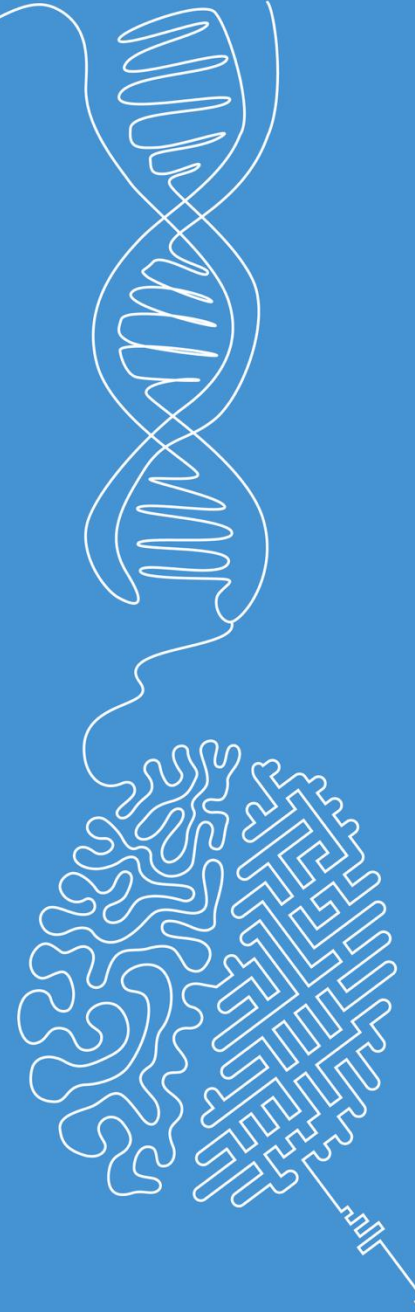


Cross-Auto-Correvolution

Image and Signal Processing

Norman Juchler



**Convolution vs.
Cross-correlation vs.
Autocorrelation**

Comparison: Convolution

■ Definitions:

- Continuous-time:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) \cdot g(t - \tau) d\tau$$

- Discrete-time:

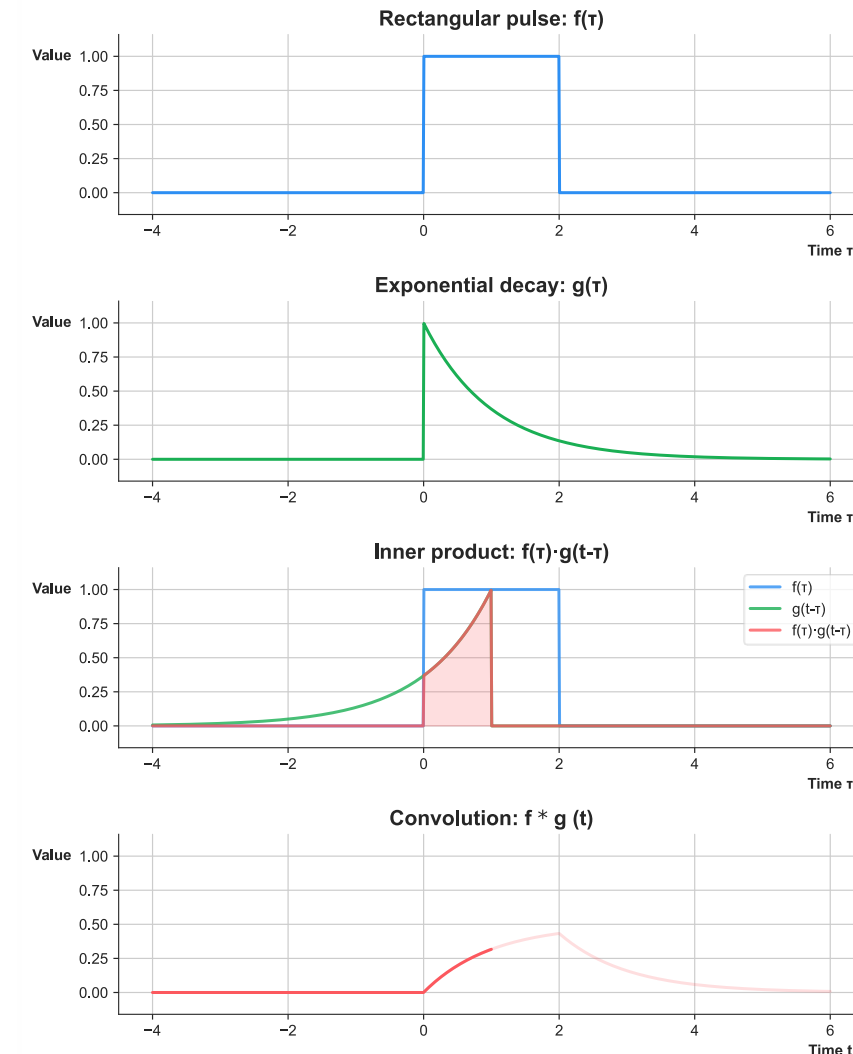
$$(f * g)[n] = \sum_{k=-\infty}^{\infty} f[k] \cdot g[n - k]$$

■ Interpretation:

- Flip one signal, shift it by t , multiply, and sum or integrate.
- Convolution can be seen as blending one signal with another.
- Describes how a system or filter transforms an input signal.
- Flip reflects the direction of cause and effect in system theory.

■ Applications:

- Audio processing: Reverb, echo, ...
- Image processing: Blurring, sharpening, edge detection, ...
- System modeling: Computing output from input via impulse response.



Comparison: Cross-correlation

Definitions:

- Continuous-time:

$$(f \star g)(t) = \int_{-\infty}^{\infty} f(\tau) \cdot g(t + \tau) d\tau$$

- Discrete-time:

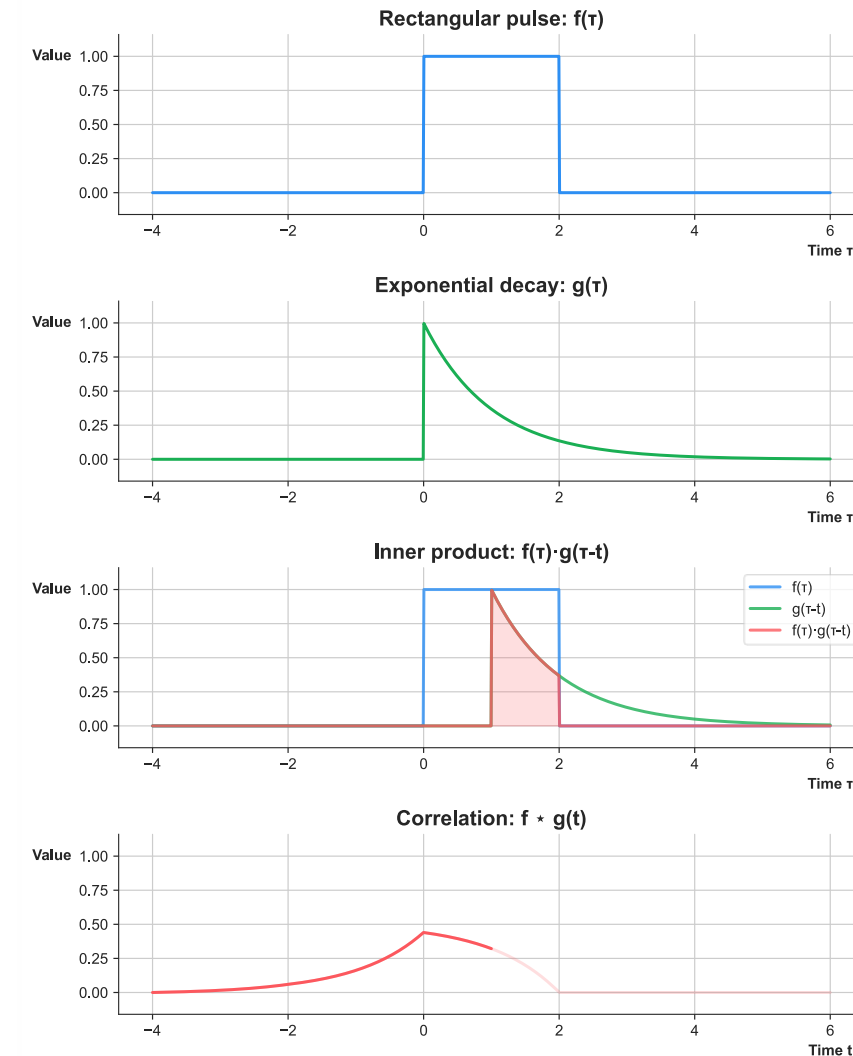
$$(f \star g)[n] = \sum_{k=-\infty}^{\infty} f[k] \cdot g[n + k]$$

Interpretation:

- Measures the similarity between two signals as one is shifted relative to the other.
- Unlike convolution, no flipping is applied.
- Peak in the output $\rightarrow$ best alignment between $f(t)$ and $g(t)$.

Applications:

- Signal synchronization: time alignment (e.g., audio video).
- Feature detection: identifying repeating patterns in time series.
- Time delay estimation between two related signals.



Comparison: Autocorrelation

■ Definitions:

- Continuous-time:

$$R_{ff}(t) = \int_{-\infty}^{\infty} f(\tau) \cdot f(t + \tau) d\tau$$

- Discrete-time:

$$R_{ff}[n] = \sum_{k=-\infty}^{\infty} f[k] \cdot f[n + k]$$

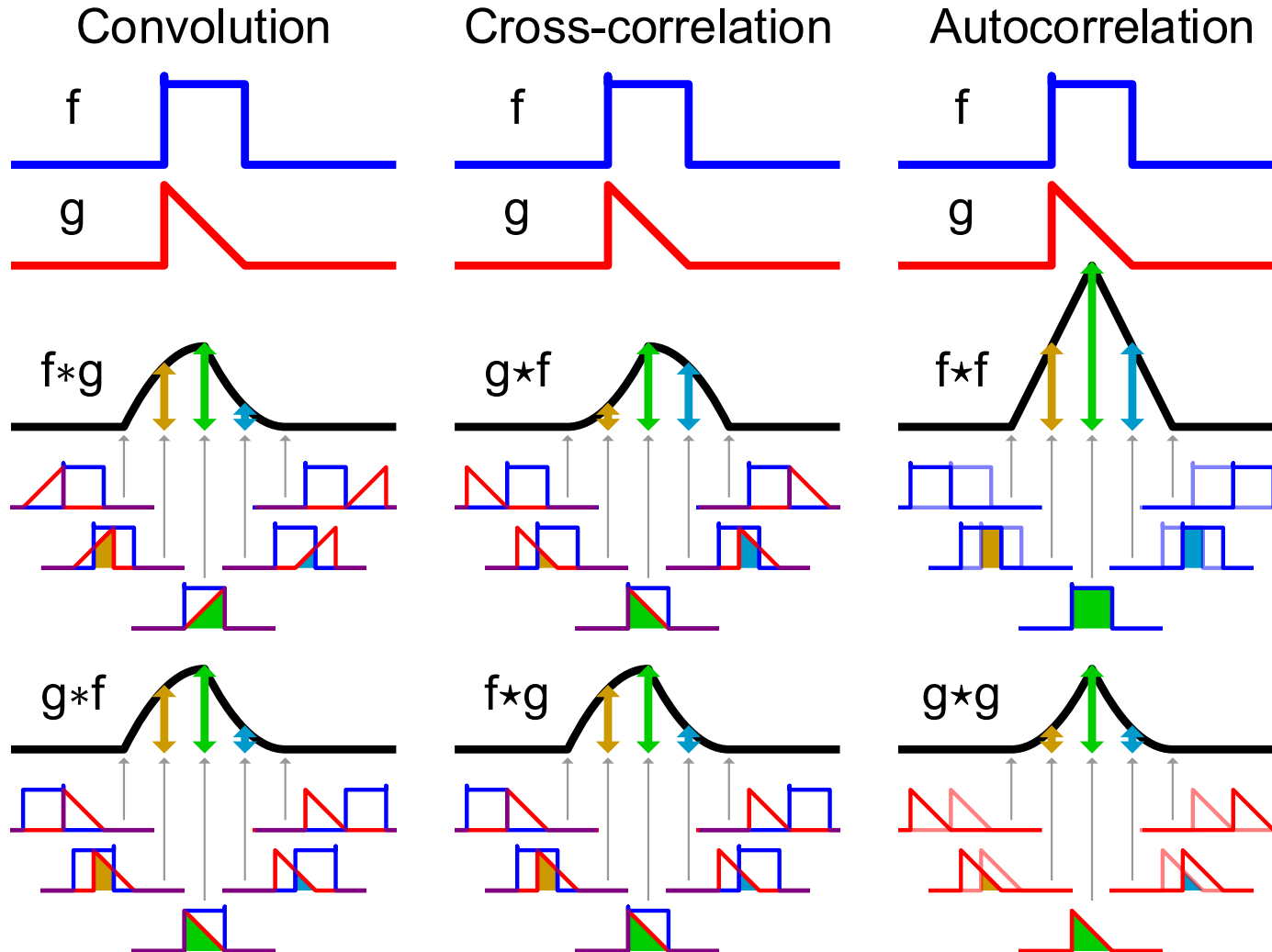
■ Interpretation:

- Autocorrelation is the cross-correlation of a signal with itself.
- Measures how much a signal resembles itself at different time shifts (lags).
- High values indicate strong self-similarity.
- Peaks reveal periodic or cyclic behavior.

■ Applications:

- Pitch detection (audio): Detecting repeating waveforms (e.g., musical notes, voiced speech).
- Pattern detection: Identifying repeated behavior in time series data.

Visual comparison



Visual comparison of convolution, cross-correlation & autocorrelation.

For the operations involving the function f , and assuming its height is 1.0, the value of the result at five different positions is illustrated by the shaded area beneath each point.

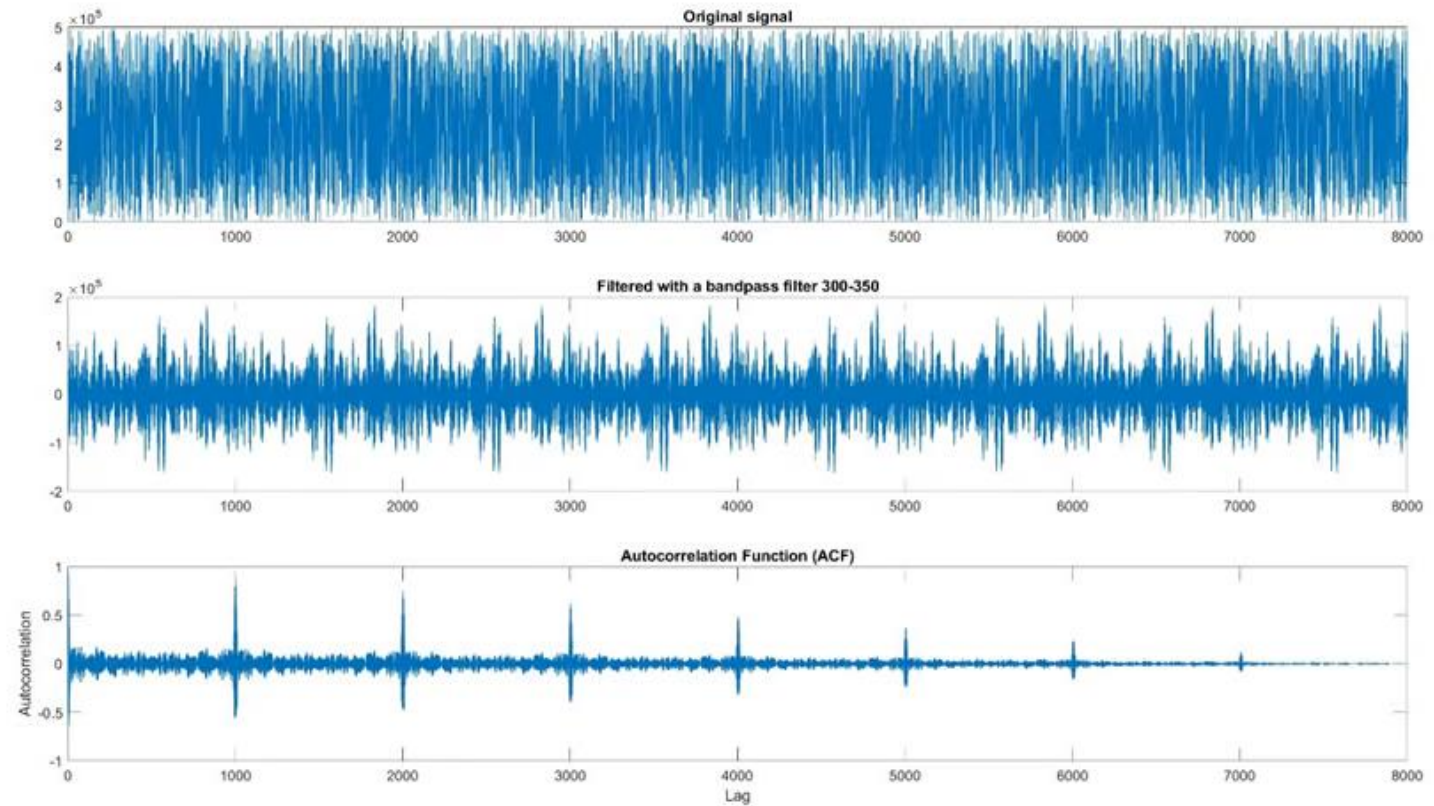
Because f is symmetric, the operations $f * g$ and $f \star g$ are identical in this example.

(Source: [Wikipedia](#))

Example: Auto-correlation

An example of using autocorrelation to identify periodicity in a signal. In this example, a very noisy signal is first processed with a bandpass filter to reduce noise, and then autocorrelation is applied to evaluate how well the signal correlates with itself at different time lags. The resulting autocorrelation plot shows that the signal repeats with a period of approximately 1000.

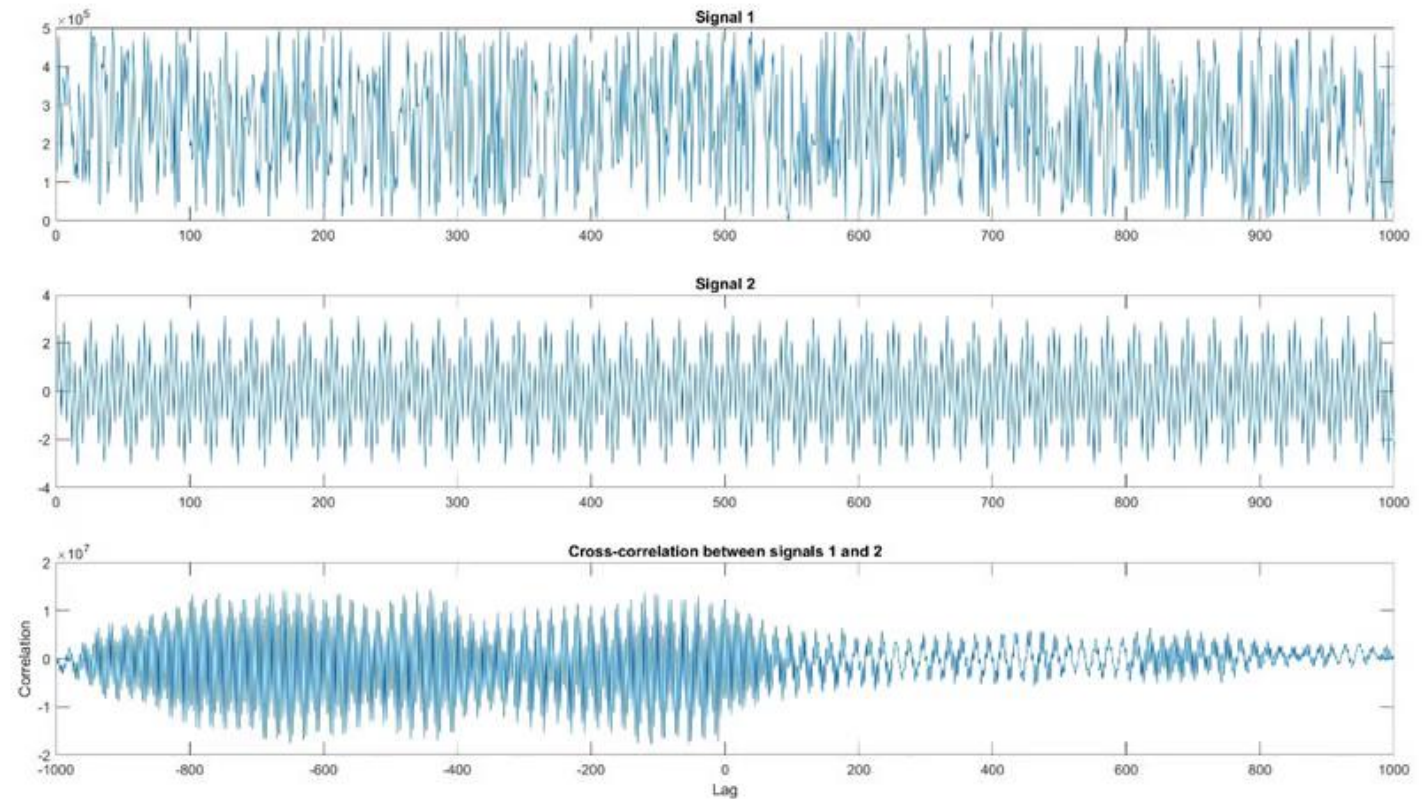
A strong autocorrelation at a specific lag indicates a repetitive pattern with that periodicity. For example, in viral infection data, autocorrelation can reveal seasonal patterns associated with outbreaks.



Example: Cross-correlation

This example illustrates the use of cross-correlation. Two signals are considered: one with weaker periodicity and another with a stronger periodic pattern. Cross-correlation is used to evaluate how well these signals align with each other at different time lags. The resulting plot shows a stronger correlation at negative lags and little to no correlation at positive lags.

Cross-correlation explores the relationship between two different signals and is useful for identifying correlations and lagged associations. For example, it can be used to study the relationship between temperature and energy consumption over time.



Further reading / watching

- [Convolution vs. Correlation / WolfSound](#)
- Geeks for Geeks: [Convolution and Cross-Correlation in CNN](#)
- Discrete Convolution: [Digital Signals Theory, Brian McFee](#)